```
"C:\Users\User 1\AppData\Local\Programs\Python\Python311\python.exe" "C:\Users\User 1\AppData\Roaming\JetBrains\PyCharmCE2023.2\scratches\MS-VECM.py"
Ticker                     BWPUSD=X  ZARUSD=X
Date
2015-01-01 00:00:00+00:00  0.106716  0.087336
2015-01-02 00:00:00+00:00  0.106229  0.086555
2015-01-05 00:00:00+00:00  0.106643  0.085268
2015-01-06 00:00:00+00:00  0.105390  0.085504
2015-01-07 00:00:00+00:00  0.105340  0.085324
...                             ...       ...
2024-11-01 00:00:00+00:00  0.075674  0.056832
2024-11-04 00:00:00+00:00  0.075729  0.056935
2024-11-05 00:00:00+00:00  0.075886  0.057135
2024-11-06 00:00:00+00:00  0.076022  0.057572
2024-11-07 00:00:00+00:00  0.075033  0.056607

[2568 rows x 2 columns]
C:\Users\User 1\AppData\Local\Programs\Python\Python311\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarning: A date index has been provided, but it has no associated frequency information and so will be ignored when e.g. forecasting.
  self._init_dates(dates, freq)
Det. terms outside the coint. relation & lagged endog. parameters for equation BWPUSD=X
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
```

```
L1.BWPUSD=X    -0.5359      0.021    -25.227      0.000      -0.578      -0.494
L1.ZARUSD=X     0.5144      0.016     32.895      0.000       0.484       0.545
L2.BWPUSD=X    -0.2094      0.019    -10.899      0.000      -0.247      -0.172
L2.ZARUSD=X     0.2749      0.018     15.369      0.000       0.240       0.310
Det. terms outside the coint. relation & lagged endog. parameters for equation
ZARUSD=X
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
L1.BWPUSD=X    -0.0192      0.030     -0.634      0.526      -0.078       0.040
L1.ZARUSD=X     0.0251      0.022      1.127      0.260      -0.019       0.069
L2.BWPUSD=X    -0.0177      0.027     -0.649      0.517      -0.071       0.036
L2.ZARUSD=X     0.0063      0.025      0.245      0.806      -0.044       0.056
              Loading coefficients (alpha) for equation BWPUSD=X
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ec1             0.0008      0.002      0.362      0.717      -0.003       0.005
              Loading coefficients (alpha) for equation ZARUSD=X
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ec1             0.0101      0.003      3.268      0.001       0.004       0.016
          Cointegration relations for loading-coefficients-column 1
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
```

```
beta.1         1.0000           0           0       0.000        1.000        1.000
beta.2        -1.3500       0.018     -77.022       0.000       -1.384       -1.316
==========================================================================================
[[-2.50072964e-04  3.58149446e-04]
 [-3.94882849e-04 -9.34650753e-05]
 [-1.69327683e-04  3.16493066e-04]
 ...
 [ 1.11636850e-04  2.11079291e-04]
 [ 1.01587607e-04  4.47394591e-04]
 [-1.16190758e-03 -9.54311966e-04]]
GMMHMM(covars_prior=array([[[-1.5, -1.5],
        [-1.5, -1.5]],

       [[-1.5, -1.5],
        [-1.5, -1.5]]]),
       covars_weight=array([[[0., 0.],
        [0., 0.]],

       [[0., 0.],
        [0., 0.]]]),
       means_prior=array([[[0., 0.],
        [0., 0.]],

       [[0., 0.],
        [0., 0.]]]),
       means_weight=array([[0., 0.],
       [0., 0.]]), n_components=2,
```

```
        n_iter=2000, n_mix=2, random_state=42,
        weights_prior=array([[1., 1.],
        [1., 1.]]))
State-dependent means (mu_{s_t}):
[[[ 0.00047952  0.00082661]
  [ 0.00010633  0.00020467]]

 [[-0.00044069 -0.00098581]
  [-0.00027787 -0.00042681]]]
State dependent Covariance matrices (Gamma_{s_t}):
[[[1.99368697e-07 2.25111198e-07]
  [6.03880116e-08 1.36891399e-07]]

 [[8.77384914e-07 6.37070857e-07]
  [7.41315553e-08 1.73490835e-07]]]
State dependent transition probabilities (alpha_{s_t}):
[[0.59327142 0.40672858]
 [0.58953317 0.41046683]]
Means of each Gaussian component in each state:
[[[ 0.00047952  0.00082661]
  [ 0.00010633  0.00020467]]

 [[-0.00044069 -0.00098581]
  [-0.00027787 -0.00042681]]]

Covariances of each Gaussian component in each state:
[[[1.99368697e-07 2.25111198e-07]
```

```
  [6.03880116e-08 1.36891399e-07]]

 [[8.77384914e-07 6.37070857e-07]
  [7.41315553e-08 1.73490835e-07]]]

Transition matrix:
[[0.59327142 0.40672858]
 [0.58953317 0.41046683]]

Initial state probabilities:
[1.00000000e+00 2.16091117e-56]

Predicted hidden states: [0 1 0 ... 0 0 1]
Likelihood Ratio Statistic: 348.74240659794305
Degrees of Freedom: 7
P-Value: 0.0
Since the p-value is 0.0, we reject the null hypothesis and conclude that the model
    with 2 components is a better fit.
Fitted Clayton copula parameter (theta): 1.101636307385832
Kendall_tau: 0.3551790726599824
Spearman: 0.5241802796774706
Lower tail dependence (lambda_L): 0.7103581453199648

Process finished with exit code 0

```